



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

## AZURE VM PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Cloud

TOTAL: 10 QUESTIONS

**Q1** What is Azure VM and what is its primary purpose in its cloud ecosystem?

Threat Model

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**Q6** How do you manage configuration and secrets for Azure VM?

Observability

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**Q2** How does IAM work with Azure VM?

Architecture

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**Q7** What are the main cost drivers for Azure VM and how do you optimize them?

Lifecycle

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**Q3** How do you monitor Azure VM in production?

Configuration

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**Q8** How do you implement high availability with Azure VM?

Compliance

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**Q4** How do you scale Azure VM to handle variable load?

Hardening

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**Q9** How does Azure VM integrate with a CI/CD pipeline?

Testing

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**Q5** How do you secure Azure VM in production?

SSO / IdP

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**Q10** What are common pitfalls when first adopting Azure VM?

Tuning

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Azure VM is a managed cloud

Mitigation

Azure VM is a managed cloud service that handles a specific infrastructure concern (compute, storage, messaging, AI). Using it shifts operational burden to the cloud provider while you focus on application logic.

A6 Store sensitive values in a managed

Observability

Store sensitive values in a managed secret store (AWS Secrets Manager, GCP Secret Manager, Azure Key Vault). Inject secrets as environment variables at runtime, never bake them into container images or source code.

A2 Access to Azure VM is controlled

Primitives

Access to Azure VM is controlled via IAM roles and policies. Attach the minimum required permissions to a service account or role. Avoid long-lived access keys; use instance roles or Workload Identity Federation instead.

A7 Cost in Azure VM typically comes

Automation

Cost in Azure VM typically comes from compute hours, data transfer, storage, and request counts. Right-size instances, use reserved capacity for steady-state workloads, and enable lifecycle policies to expire old data.

A3 Azure VM emits metrics to the

Hardening

Azure VM emits metrics to the cloud provider's monitoring service (CloudWatch, Cloud Monitoring, Azure Monitor). Set alerts on key metrics (error rate, latency, capacity). Use distributed tracing to correlate across services.

A8 Deploy Azure VM across multiple availability

Governance

Deploy Azure VM across multiple availability zones. Use managed replicas or active-active configurations. Implement health checks and automatic failover. Test resilience regularly with chaos engineering or failover drills.

A4 Azure VM supports auto-scaling policies based

Gotchas

Azure VM supports auto-scaling policies based on CPU, queue depth, or custom metrics. Set minimum and maximum capacity. Horizontal scaling (more instances) is generally preferred over vertical for cloud workloads.

A9 CI/CD pipelines use the cloud CLI

Assessment

CI/CD pipelines use the cloud CLI or SDK to deploy updates to Azure VM after tests pass. Infrastructure-as-code (Terraform, CDK) defines Azure VM configuration as version-controlled code deployed via the pipeline.

A5 Place Azure VM in a private

Federation

Place Azure VM in a private subnet with no public endpoint where possible. Use VPC security groups or firewall rules to allow only required traffic. Encrypt data at rest and in transit. Rotate credentials regularly.

A10 Common mistakes include misconfigured IAM

Optimization

Common mistakes include misconfigured IAM policies (too permissive), no cost alerting, deploying only in one availability zone, and missing backups. Read the service SLA carefully to understand what the cloud provider guarantees.